

Combinatorics – 3

Combinatorial numbers: Fibonacci, Catalan, partitions

Homework

14 сентября 2026 года

Theoretical minimum. $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$; $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, whence Cassini's identity $F_{n-1}F_{n+1} - F_n^2 = (-1)^n$, the addition formula $F_{m+n} = F_mF_{n+1} + F_{m-1}F_n$ and $\gcd(F_m, F_n) = F_{\gcd(m,n)}$; also $\sum_k \binom{n-k}{k} = F_{n+1}$, and every positive integer is uniquely a sum of non-consecutive Fibonacci numbers (Zeckendorf). The Catalan numbers $C_0 = 1$, $C_{n+1} = \sum_k C_k C_{n-k}$ count Dyck paths, balanced brackets, triangulations of a convex polygon, binary trees and ballot sequences, and $C_n = \frac{1}{n+1} \binom{2n}{n}$ by the cycle lemma. For partitions $\sum p(n)x^n = \prod_k (1-x^k)^{-1}$, conjugation of Young diagrams exchanges the number of parts with the largest part, and Euler's pentagonal number theorem gives a fast recurrence. Finally Vandermonde's identity and Lucas' theorem modulo a prime. Closed forms for F_n and C_n by generating functions are in Functional sequences, series and spaces – 3, homework 9.

1. In the Euclidean plane a hopper starts at the origin; at the n -th minute it hops a distance F_n north, south, east or west. Prove that if the first two hops were perpendicular, the hopper can never return to its starting point. (The Putnam problem on the polynomial of degree 1008 interpolating the Fibonacci numbers is Polynomials – 7, problem 2.) (ICMC)
2. Prove that 3^{2023} divides $3^2 F_4 + 3^3 F_6 + 3^4 F_8 + \dots + 3^{2023} F_{4046}$. (ICMC)
3. Prove Euler's pentagonal number theorem $\prod_{k \geq 1} (1-x^k) = \sum_{j \in \mathbb{Z}} (-1)^j x^{j(3j-1)/2}$ by Franklin's involution on Young diagrams, and turn it into a recurrence computing $p(n)$ in $O(n^{3/2})$ steps.

4. Prove the identities $\sum_k \binom{n-k}{k} = F_{n+1}$, $\sum_k \binom{n}{k} F_k = F_{2n}$ and $\sum_k (-1)^k \binom{n}{k} \binom{2n-2k}{n} = 2^n$, each by interpreting both sides as a count or by extracting a coefficient. (Vandermonde's identity and $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$ are proved by double counting in Methods of proof – 4, homework 4.)
5. Given a positive integer n , let $M(n)$ be the largest integer m for which $\binom{m}{n-1} > \binom{m-1}{n}$. Evaluate $\lim_{n \rightarrow \infty} M(n)/n$. (Putnam)
6. Prove Lucas' theorem for $\binom{n}{k}$ modulo a prime p , deduce that the number of odd binomial coefficients in row n of Pascal's triangle is $2^{s(n)}$ with $s(n)$ the number of ones in the binary expansion of n , and find all n for which every $\binom{n}{k}$, $0 < k < n$, is even.
7. Define the Stirling numbers $S(n, k)$ and $c(n, k)$ and prove $x^n = \sum_k S(n, k) x(x-1) \cdots (x-k+1)$ and $\sum_k c(n, k) x^k = x(x+1) \cdots (x+n-1)$. Deduce the recurrence for the Bell numbers $B_{n+1} = \sum_k \binom{n}{k} B_k$.
8. Prove that the number of partitions of n into parts of size at most k equals the number of partitions of n into at most k parts, by conjugating Young diagrams, and prove that the number of self-conjugate partitions of n equals the number of partitions into distinct odd parts.
9. Prove that F_{n+1}/F_n is the n -th convergent of the continued fraction $[1; 1, 1, \dots]$, deduce $|F_{n+1}/F_n - \varphi| < 1/F_n^2$, and explain the connection with the losing positions of Fibonacci nim (Game Theory – 1, homework 4).
10. Prove the Catalan convolution identity $\sum_k C_k C_{n-k} = C_{n+1}$ bijectively, and prove that the number of ways to triangulate a convex $(n+2)$ -gon by non-crossing diagonals equals C_n , by induction on the triangle containing a fixed side.